Hein Zarni Aung

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EDUCATION

Stony Brook University **Expected Graduation: May 2027**

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Relevant Coursework: Full Stack Development, Software Security, Systems Fundamentals I & II, Algorithms

RESEARCH

Undergraduate Research Assistant — Ethos Lab, SBU **Sep 2025 – Dec 2025 (paused)**

PI: Abisheka Pitumpe | Team size: 2

- Built a pipeline to monitor newly issued SSL/TLS certificates in real time and flag potentially fraudulent domains impersonating EZ-Pass tolling services, scraping keywords from the raw HTML, CSS, JS, and URLs of newly created sites.
- Used Selenium to automate certificate and domain scraping and Python to filter and triage flagged sites; auto-captured homepage screenshots of promising candidates to speed up manual review.
- Collaborated in a two-person team with one honors student under a faculty PI to design keyword- and heuristic-based detection logic for scam website identification.

INTERNSHIPS

Software Engineering Intern — Clustr **Jan 2026 – Sept 2026 (expected)**

Cross-platform React Native social/scheduling app | ~8-person dev team

- Contributed to Clustr, a cross-platform React Native social/scheduling app, delivering 50+ commits across backend and frontend features.
- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows, like/dislike interactions, and activity feed endpoints.
- Implemented secure authentication features, including a password reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.
- Collaborated with an 8-person, multi-developer team via Git, regularly merging and resolving conflicts across feature branches to keep community and scheduling modules in sync with the shared codebase.

Software Engineering Intern — Prenora Dynamics LLC **Jun 2026 – Sept 2026 (expected)**

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).

PROJECTS

TuneAcademy — Hack Brooklyn 2026 (36-hour hackathon) **Apr 2026**

Winner: Best Prenora Dynamics Hack (Sponsor Challenge) | Team of 4

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time and compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Collaborated in a 4-person team; broader app included a React/Tailwind frontend, Python/FastAPI backend in Docker, Cloud Firestore, GCP hosting, and ElevenLabs-generated voice feedback reports.

Discord Voice Channel Activity Monitor (Node.js, Discord.js, Cloud)

- Built a Node.js/Discord.js bot to solve a scheduling problem for a friend group spread across different schedules, notifying users in real time when someone joins a voice channel so they can catch impromptu hangouts or gaming sessions.
- Implemented customizable DM notifications, VIP lists, and per-server/per-user channel thresholds with JSON-based configuration.
- Deployed on a cloud server using PM2; reached 50 concurrent active users.

CAMPUS INVOLVEMENT

Teaching Assistant – Data Structures & Algorithms **Aug 2025 – Dec 2025**

Dept. of Computer Science, Stony Brook University

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original PowerPoint-based review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C
Frontend: React, React Native, Next.js
Backend: Node.js, Express.js, FastAPI
Databases & Cloud: MongoDB, PostgreSQL, Firebase/Firestore, AWS, GCP
Tools: Git, Docker, Linux, PM2, Selenium